

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801D - MAINTENANCE AND REPAIR OF CONCRETE STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The primary reason corrosion is critical in RCC structures is because it
 - a) reduces workability of concrete
 - b) decreases alkalinity of cement paste
 - c) leads to loss of steel cross-section and bond
 - d) increases creep strain
 - 2 The passive layer on steel in concrete is maintained due to
 - a) low moisture content
 - b) high alkalinity of pore solution
 - c) high cement content
 - d) presence of aggregates
 - 3 Which factor accelerates corrosion most significantly?
 - a) High cement content
 - b) Increased cover thickness
 - c) Presence of chlorides
 - d) Low water–cement ratio
 - 4 Chloride-induced corrosion primarily affects
 - a) aggregates
 - b) cement paste
 - c) steel reinforcement
 - d) formwork
 - 5 Which zone acts as cathode in corrosion cell?
 - a) Area with higher oxygen availability
 - b) Area with lower moisture
 - c) Area with higher chloride content
 - d) Area with cracks
 - 6 Formation of ettringite is associated with
 - a) carbonation
 - b) sulphate attack
 - c) acid attack
 - d) ASR

- 7 Which sulphate is most aggressive to concrete?
 - a) Calcium sulphate
 - b) Magnesium sulphate
 - c) Sodium sulphate
 - d) Potassium sulphate

- 8 Acid-resistant concrete mainly requires
 - a) high C_3A content
 - b) dense and low permeability mix
 - c) higher water–cement ratio
 - d) increased fine aggregate

- 9 Alkali-Silica Reaction requires
 - a) reactive aggregates, moisture, alkalis
 - b) sulphates and acids
 - c) carbonation and oxygen
 - d) chlorides only

- 10 ASR leads to formation of
 - a) calcium hydroxide
 - b) expansive gel
 - c) ettringite
 - d) gypsum

- 11 Ultrasonic Pulse Velocity test helps in assessing
 - a) surface roughness
 - b) density and quality of concrete
 - c) chloride content
 - d) reinforcement diameter

- 12 Low UPV value generally indicates
 - a) good quality concrete
 - b) dense concrete
 - c) poor quality or cracked concrete
 - d) high strength concrete

- Q2. Solve the following.
 - A) Explain embedded metal corrosion in concrete structures. 6
 - B) Explain the role of alkalinity of concrete in protecting reinforcement. 6

- Q3. Solve the following.
 - A) Explain the process of deterioration due to Alkali-Silica Reaction. 6
 - B) Explain frost attack and suggest preventive measures. 6

- Q4. Solve Any Two of the following.
 - A) Explain in detail the rebound hammer test. 6
 - B) Explain the necessity of maintenance activities in structural operation. 6
 - C) What is ground penetrating radar system and how it detects delamination? 6

- Q5. Solve Any Two of the following.
 - A) Explain the different steps of assessment of condition of RCC structures. 6
 - B) Explain cosmetic and structural surface repairs. 6
 - C) How to conduct visual inspection of concrete structures? 6

Q6. Solve Any Two of the following.

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| A) Explain injection grout method for concrete repair. | 6 |
| B) Explain the process of strengthening of slabs and beams. | 6 |
| C) Explain external post-tensioning repair method. | 6 |

***** End *****